CO-2 Assessment Tool - 1

V. Venkata Susheel Chowdary
192472060

Analytical Problem Solving

1. Thresholding and intensity-level slicing - PCB inspection.

Given,

original pixel value

$r = [25, 60, 95, 130, 165, 200, 235]$

solder-joint intensity band : 90-170

(a) Binary thresholding

$$s = \begin{cases} 255 & r \geq 128 \\ 0 & r < 128 \end{cases}$$

original r	output s
25	0
60	0
95	0
130	255
165	255
200	255
235	255

(b) intensity-level slicing

Rule : Set pixels in the range 90-120 to 255 ; leave all other pixels unchanged.

Original r	Output s
25	25
60	60
95	255
130	255
165	255
200	200
235	255

© Number and percentage of highlighted pixel

95, 130 and 165

Total pixel = 7

$$\text{percentage} = \frac{3}{7} \times 100$$

$$= 42.86$$

(d) Original mean:

$$\bar{r} = \frac{25+60+95+130+165+200+235}{7}$$

$$= \frac{910}{7} = 130$$

$$\bar{s} = \frac{25+60+255+255+255+200+235}{7}$$

$$= \frac{1285}{7} = 183.57$$

(e) Intensity level slicing is more appropriate because it highlight only the pixels within the required solder-joint bond at 90-120. Binary thresholding also highlights pixels 200 and 235 which lie outside the desired bond.

2. Piecewise-linear contrast stretching - x-Ray image enhancement.

*> L=255, output range 0-255

*> Control point; $(r_1, s_1) = (70, 20)$

$(r_2, s_2) = (140, 230)$

input pixels : [30, 70, 80, 110, 140, 200]

@ For a straight line

$$s = ar + b$$

where a is the slope

segment 1: $0 \le r \le 70$ (0,0) and (70,20)

$$a_1 = \frac{20-0}{70-0} = \frac{2}{7} = 0.2857$$

$$s_1 = \frac{2}{7} r$$

segment 2: $70 \le r \le 140$

$$a_2 = \frac{230-20}{140-70} = \frac{210}{70} = 3$$

$$s = ar + b$$

$$20 = 3(70) + b \Rightarrow b = -190$$

$$s_2 = 3r - 100$$

Segment 3: $140 \le r \le 255$

$$a_3 = \frac{255-230}{255-140} = \frac{25}{115} = 0.2174$$

$$s_3 = 230 + 0.2134 (r - 140)$$

$$s_3 = 230 + \frac{5}{23} (r - 140)$$

(6) Calculate transformed value

input r	equation used	transformed s
30	$2r/7$	8.57
70	$2r/7$	20
90	$3r - 190$	80
110	$3r - 190$	140
140	$3r - 190$	230
200	$230 + \frac{5}{23}(r - 140)$	243.04

(b)

(c) Band width and contrast gain

$W_{original} = 140 - 70 = 70$

Transformed band width:

$W_{Transformed} = 230 - 20 = 210$

gain factor

(c)

$$G = \frac{W_{trans}}{W_{org}} = \frac{210}{70} = 3 //$$

(d)

Segment	Slope	Effect
0-70	0.2857	compression
70-140	3	Expansion
140-255	0.2174	Compression

(d)

(e) The transformation improves visibility of diagnastic details b/w gray level 70 and 140 because their range expands from 70 levels to 210 levels.

3. Spatial Filtering- impulse Noise Removal.

$$\begin{bmatrix} 52 & 55 & 58 \\ 54 & 255 & 57 \\ 53 & 56 & 59 \end{bmatrix}$$

(e)

(a) 3×3 mean filter

$$\text{mean} = \frac{\text{sum of all 9 pixels}}{9}$$

$$= \frac{52+55+58+54+255+57+53+56+59}{9}$$

$$= 77.67$$

(b) weighted - average filter

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$= \frac{1}{16}\left[(52+58+53+59)+2(55+54+57+56)+4(255)\right]$$

$$= 105.375$$

(c) Arrange all nine pixel values in ascending order

52, 53, 54, 55, 56, 57, 58, 59, 255

median = 56.

(d) Absolute error of each filter

$$R = 52+55+58+54+57+53+56+59/8$$

$$= 55.5$$

Absolute error formula

$$E = |\text{Filter output} - R|$$

$$E_{mean} = |77.67 - 55.5| = 22.17$$

$$E_{weighted} = 49.815$$

$$E_{median} = |56 - 55.5| = 0.5$$

(e) The median filter is the most approriate it removes the extreme salt-noise value of 255 & produces the smallest error ; 0.5. It is also generally effective for salt and pepper noise.

4. Edge detection - sobel and prewitt Gradient Analysis.

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 90 \\ 10 & 90 & 90 \end{bmatrix}$$

(a) Sobel G_x and G_y

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \qquad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

$$G_x = (-10+10) + (-20+180) + (-10+90)$$

$$G_x = 240$$

$$G_y = (-10-20-10) + (10+180+90)$$

$$= 240$$

(b) Gradient

$$m = \sqrt{G_x^2 + G_y^2}$$

$$m = 339.41$$

$$m_{app} = |G_x| + |G_y|$$

$$= 480$$

$$\text{error} = \frac{|480 - 339.41|}{339.41} \times 100$$

$$= 41.42\%$$

(c) Gradient

$$\alpha = \tan^{-1}\left(\frac{G_x}{G_y}\right) = \tan^{-1}(1) = 45^\circ$$

(d) Prewitt operation

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} \qquad G_y = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$G_x = (-10+10) + (-10+90) + (-10+90)$$

$$G_x = 160$$

$$G_y = 160$$

$$m = \sqrt{160^2 + 160^2}$$

$$= 226.27$$

$$m_{app} = 320$$

Percentage error $= \frac{320 - 226.27}{226.27} \times 100$

$= 41.42\%$

© edge detection for $m \geq 250$

operator	exact magnitude	threshold	edge detected?
Sobel	339.41	250	Yes
Prewitt	226.27	250	No.

5. Global Thresholding - Iterative and otsu segmentation.

intensity r_k :	0	1	2	3	4	5	6	7
n_k :	3	6	4	1	1	5	8	4

Total no. of pixels:

$$N = 3+6+4+1+1+5+8+4 = 32$$

(a) global mean intensity

$$m_{C_7} = \frac{\sum r_k n_k}{N}$$

$$M_G = \frac{0(3) + 1(6) + 2(4) + 3(1) + 4(1) + 5(5)}{32}$$

$$= 3.8125$$

(b) Initial threshold

$$T_0 = m_{C_7} = 3.8125$$

I_1

Class 1 : intensity 0-3

Class 2 : intensity 4-7

$$m_1 = \frac{0(3) + 1(6) + 2(4) + 3(1)}{3+6+4+1} = \frac{17}{14} = 1.2143$$

$T_1 = \frac{m_1 + m_2}{2} = 3.5238$

$\Delta T = |3.5238 - 3.8125| = 0.2887$

$T = 3.5238 = 3.52$

(c) otsu method

$k = 2$

$\sigma_B^2(k) = P_1(k) P_2(k) [m_1(k) - m_2(k)]^2$

$P_1 = \frac{13}{32} = 0.40625$ $\quad P_2 = \frac{19}{32} = 0.59375$

$m_1 = 14/13 = 1.0769$ $\quad m_2 = 108/12 = 5.6842$

$\sigma_B^2(2) = 5.1202$

(d) optimum threshold and binary mapping

Threshold k	Between – class variance
2	5.1207
3	5.2506
4	5.1354

max. variance occurs $k = 3$.

The no. of objects = 14

Percentage = $\frac{14}{32} \times 100 = 43.75\%$.

(e)

Method	Threshold	Result
iterative thresholding	3.5238	seperates levels 0-3 from 4-7
otsu's method	3	seperates levels 0-3 from 4-7